MEDIUM · ALGEBRA

Algebra

04

10 questions · ~15 min

Q1 ALGEBRA

In the xy-plane, line k passes through the points $0, -5$ and $1, -1$. Which equation defines line k?

A. $y = -x + \frac{1}{4}$

B. $y = \frac{1}{4}x - 5$

C. $y = -x + 4$

D. $y = 4x - 5$

Q2

GRID-IN ALGEBRA

$$\frac{1}{3}x + 6 - \frac{1}{2}x + 6 = -8$$

What value of x is the solution to the given equation?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ALGEBRA

According to a model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to four times the body weight of the bee, in grams. According to the model, what would be the head width, in millimeters, of a worker bumblebee that has a body weight of 0.5 grams?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

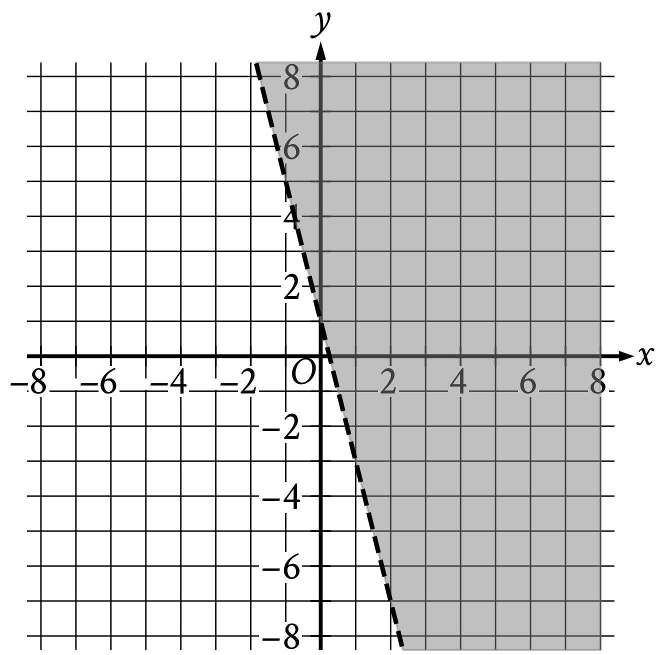
Two customers purchased the same kind of bread and eggs at a store. The first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. The second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. What is the cost, in dollars, of 1 dozen eggs?

A. 3.77

B. 3.88

C. 4.15

D. 4.34



- The boundary of the inequality is a dashed line.
    - The line slants sharply down from left to right.
    - The line passes through the following points:
        - (negative 1 comma 5)
        - (0 comma 1)
        - (1 comma negative 3)
- The area above and to the right of the boundary is shaded.

The shaded region shown represents the solutions to which inequality?

A. $y < 1 + 4x$

B. $y < 1 - 4x$

C. $y > 1 + 4x$

D. $y > 1 - 4x$

Q6

ALGEBRA

In an article about exercise, it is estimated that a 160-pound adult uses 200 calories for every 30 minutes of hiking and 150 calories for every 30 minutes of bicycling. An adult who weighs 160 pounds has completed 1 hour of bicycling. Based on the article, how many hours should the adult hike to use a total of 1,900 calories from bicycling and hiking?

A. 9.5

B. 8.75

C. 6

D. 4

Q7

GRID-IN ALGEBRA

$$2x + 16 = ax + 8$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

ALGEBRA

The function h is defined by $hx = 4x + 28$. The graph of $y = hx$ in the xy-plane has an x-intercept at $a, 0$ and a y-intercept at $0, b$, where a and b are constants. What is the value of $a + b$?

A. 21

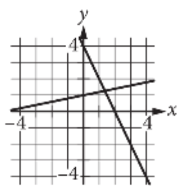
B. 28

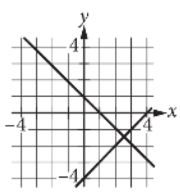
C. 32

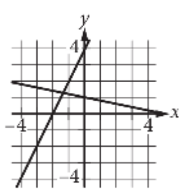
D. 35

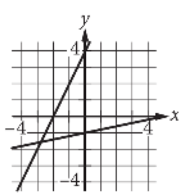
$$x + 5y = 5$$
$$2x - y = -4$$

Which of the following graphs in the xy-plane could be used to solve the system of equations above?

A. 

B. 

C. 

D. 

Q10

GRID-IN ALGEBRA

An event planner is planning a party. It costs the event planner a onetime fee of $\$35$ to rent the venue and $\$10.25$ per attendee. The event planner has a budget of $\$200$. What is the greatest number of attendees possible without exceeding the budget?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.